social networks, media services, data analytics, and cloud platforms, simply termed as SMAC (social, mobile, analytics and cloud). Internet of Humans deals with the smart devices that interact with humans like smart wristbands and connected medical devices to monitor and collect various datasets. These can be used for remote patient monitoring using cloud-based platforms or remote health advisory solutions through contact centre AI services.

Internet of Everything (IoE) is an emerging technology and requires costly infrastructure. However, smart ready-to-use IoE services are the future.

Some popular real-time use cases of Internet of Everything (IoE)

Here are the top 10 popular use cases of Internet of Everything (IoE).

Smart cities

- **Real-time traffic management**: IoE sensors monitor traffic flow, optimise traffic lights, and reroute vehicles to avoid congestion, reducing travel times and emissions.
- **Environmental monitoring**: Air quality sensors track pollutants in real-time, allowing for immediate adjustments and promoting public health.
- **Smart waste management**: Connected trash bins indicate fill levels, enabling efficient waste collection and reducing unnecessary truck trips.

Industrial automation

- **Predictive maintenance**: Sensors on industrial equipment monitor performance and predict potential failures in real-time. This allows for preventive maintenance, avoiding costly downtime and ensuring smooth operations.
- **Remote monitoring and control**: Industrial processes can be monitored and controlled remotely via IoE, improving safety and allowing for adjustments based on real-time data.
- **Real-time inventory tracking**: IoE tracks inventory levels, optimising stock management and preventing stock outs or overstocking.

Energy management

- **Smart grids**: IoE enables two-way communication between power grids and consumers. Real-time data on energy usage allows for dynamic pricing, optimised distribution, and integration of renewable energy sources.
- **Demand response management**: Consumers can adjust energy usage based on real-time grid demands, promoting efficient energy consumption and grid stability.
- **Smart homes and buildings**: IoE thermostats and appliances adjust energy consumption based on real-time occupancy and weather data, reducing energy waste and lowering utility bills.

Healthcare

- **Remote patient monitoring**: Wearable devices and sensors track vital signs like heart rate, blood pressure, and blood sugar in real-time, enabling remote patient monitoring and timely intervention during emergencies.
- **Connected medical devices**: IoE allows medical devices to share data in real-time, improving diagnosis, treatment, and patient outcomes.
- **Hospital resource management**: IoE can track location and usage of medical equipment and supplies, optimising resource allocation and improving hospital efficiency.

Retail and supply chain management

- **Real-time asset tracking**: IoE tracks the location and condition of goods throughout the supply chain, improving transparency, reducing loss, and optimising delivery routes.
- **Personalised marketing**: IoE can gather real-time customer data in stores, enabling targeted promotions and enhancing the overall shopping experience.
- **Inventory management**: Real-time tracking of inventory levels allows retailers to optimise stock levels and prevent stockouts, leading to improved customer satisfaction.

Agriculture

- **Precision agriculture**: Sensors monitor soil moisture, nutrients, and weather conditions in real-time, allowing farmers to optimise irrigation, fertilisation, and other practices, increasing yield and reducing waste.
- **Livestock monitoring**: IoE sensors can track the health and location of livestock, enabling early detection of disease and improving animal welfare.
- **Real-time crop monitoring**: IoE allows farmers to monitor crop health and growth remotely, assisting in making informed decisions about harvesting and resource allocation.

Environmental monitoring

- **Forest fire detection**: IoE sensors can detect fires in real-time, enabling faster response and minimising damage.
- **Water quality monitoring**: Sensors in water bodies can monitor water quality in real-time, allowing for timely intervention in case of pollution.
- **Flood prediction and monitoring**: IoE can collect and analyse data from various sources to predict and monitor floods, enabling early warnings and evacuation efforts.

Connected cars

- **Real-time traffic updates**: Connected cars share traffic data, enabling navigation systems to provide real-time updates and suggest alternative routes.